Online E-Commerce System for Sales and Order Management

BY

Avishek Bista

T.U. Registration No./ T.U. Roll No.:7-2-282-16-2022 | 14892/22

College Roll No.:022BIM016

St. Xavier's College

A Project Report Submitted to

Faculty of Management, Tribhuvan University

in partial fulfillment of the requirements for the degree of

Bachelor of Information Management (BIM)

Maitighar, Kathmandu

January, 2025

STUDENT DECLARATION

This is to certify that I have completed the Project entitled **"Online E-Commerce System for Sales and Order Management"** under the guidance of **"Mr. Pravin Ratna Shakya"** in partial fulfillment of the requirements for the degree of Bachelor of Information Management at Faculty of Management, Tribhuvan University. This is my original work, and I have not submitted it earlier elsewhere.

Date: Signature:

Name:

SUPERVISOR RECOMMENDATION

This is to certify that the project entitled "**Online E-Commerce System for Sales and Order Management**" is an academic work done by "**Avishek Bista**" submitted in the partial fulfillment of the requirements for the degree of Bachelor of Information Management at Faculty of Management, Tribhuvan University under my guidance and supervision. To the best of my knowledge, the information presented by him/her in the project report has not submitted earlier.

_____

Signature of the Supervisor

Name:

Designation:

Date:

APPROVAL SHEET

This is to certify that the project titled **Online E-Commerce System for Sales and Order Management** submitted by **Avishek Bista** has been examined and approved. In our opinion, it meets the required scope and quality standards for a project submitted in partial fulfillment of the requirements for the degree of Bachelor of Information Management (BIM).

Approval Panel:

S.No.	Name	Designation	Signature
1	Mr. Pravin Ratna Shakya	Project Supervisor	
2	Mr. Ganesh Yogi	Program Coordinator	
3	Mr. Rameshwor Suwal	Internal Examiner	
4		External Examiner	
5		External Examiner	

Date of Defense:……..

Department: Department of Computer Science

Faculty: Faculty of Management

ACKNOWLEDGEMENT

It is my pleasure to use this occasion to appreciate my project supervisor, **Mr. Pravin Ratna Shakya,** the frequent guidance, constructive suggestions and feedback he provided me throughout the development of my current project which is referred to as Online E-Commerce System in Sales and Order Management. Their help and professional guidance were quite useful to me to achieve success in this work.

I want to thank **Mr. Ganesh Yogi,** the head of Department, from the bottom of my heart for his guidance and support, which have made both my academic journey and development of my project much better.

I would also like to sincerely thank our internal examiner **Mr. Rameshwor Suwal,** for his valuable time, constructive feedback and support during the evaluation of the project work

I also owe the Faculty of Management, Tribhuvan University, and St. Xavier to provide me with the academic environment, materials, and the potential to carry out such project as partial completion of the Bachelor of Information Management (BIM) degree.

I would also like to thank my teachers and other employees who in any form gave the project knowledge, support, and encouragement.

The discussions, feedback and motivation were also provided and assisted by my friends and classmates, and I would like to thank them as well. And finally, I would like to give credit to the fact that thanks to my family the family has always stood by me, has been patient and encouraging to me in my academic life and for doing this project.

Avishek Bista
14892/22

ABSTRACT

This is an online e-commerce system in terms of sales and order. Management" proposes elaboration and development of the web-based platform. that automates the sales and order taking process and the transactions. The conventional selling approaches are typically found in small and medium businesses. which may lead to manual records keeping or unofficial social media transactions. data inconsistency, inefficient monitoring and ineffectiveness.

The proposed system is a web-based system that provides a centralized solution where customer can. register, log in, shop, work with shopping carts and order. online. Administrators can administer products, viewing and tracking orders. logs of the sales through a special dashboard. The system is developed implementing the modern web technologies on the Django platform on the back end, and data storage Structured data is stored through a relational database.

The methodology to be applied in the development of this project is the Iterative Waterfall that involves the following. implementation, system design, requirement analysis, testing and deployment. planning. It is largely a Transaction Processing System.

(TPS), ensuring that sales and order data must be input correctly and on-demand and providing a point of further development into reporting and analytics. features. The results of the testing have shown that the system is dependable within the key modules. like authentication, managing carts, processing and transaction of orders. recording. The application developed is efficient in terms of operations and data. Validity, and usefulness of e-sales management.

TABLE OF CONTENTS

TITLE PAGE ..... *i*
STUDENT DECLARATION ..... *ii*
SUPERVISOR'S RECOMMENDATION ..... *iii*
APPROVAL SHEET ..... *iv*
ACKNOWLEDGEMENT ..... *v*
ABSTRACT ..... *vi*
TABLE OF CONTENTS ..... *vii*
LIST OF FIGURES ..... *ix*
LIST OF TABLES ..... *x*
LIST OF ABBREVIATIONS ..... *xi*
CHAPTER I: INTRODUCTION 1
1.1 Background of the study 1
1.2 Problem Statement 2
1.3 Objectives of the Project 2
1.4 Literature Review 2
1.4.1 Overview of Existing Systems 2
1.4.2 Technologies Used in Similar Systems 2
1.4.3 Research and Academic Works 3
1.5 Development Methodology 6
1.6 Scope and Limitation of the Project 7
1.7 Report Organization 7
CHAPTER II: SYSTEM DEVELOPMENT PROCESS 9
2.1 Analysis 9
2.1.1 Requirement Analysis 9
2.1.2 Feasibility Study 11
2.1.3 System Requirements Specification (SRS) 12
2.2 System Design 12
2.2.1 System Architecture 12
2.2.2 Database Design 13
2.2.3 User Interface Design 14
2.2.4 System Flowchart 20
2.2.5 Object-Oriented Modelling 23

2.3 Implementation 26
2.3.1 Tools and Development Environment 26
2.3.2 System Modules and their Implementation 27
2.3.3 Coding Standards and Conventions 28
2.3.4 Security Considerations 28
2.3.5 Testing and Debugging 28
2.4 Deployment 29
2.4.1 Local Deployment Configuration 29
2.4.2 Planned Online Deployment 29
CHAPTER III: CONCLUSION AND RECOMMENDATIONS 30
3.1 Summary 30
3.2 Conclusion 30
3.3 Recommendations 31
REFERENCES 31
APPENDICES 33

LIST OF FIGURES

Figure 1.1: Iterative Waterfall Model for E-Commerce System 6
Figure 2.1: Entity Relationship Diagram 13
Figure 2.2: Admin Dashboard 14
Figure 2.3: User Dashboard 14
Figure 2.4: Sign In Page 15
Figure 2.5: Log In Page 15
Figure 2.6: Home Page 16
Figure 2.7: Product Listing Page 17
Figure 2.8: Product Detail Page 18
Figure 2.9: Cart Page 18
Figure 2.10: Checkout Page 19
Figure 2.11: Flowchart of Online E-Commerce System 20
Figure 2.12: Use Case Diagram of Online E-Commerce System 22
Figure 2.13: Class Diagram of Online E-Commerce System 23
Figure 2.14: Object Diagram of Online E-Commerce System 24
Figure A1: Homepage of the Website 33
Figure A2: Product Listing Page 33
Figure A3: About Us Page 34
Figure A4: Contact Us Page 34
Figure A5: Cart and Checkout Page 34
Figure A6: Sign In Page 35
Figure A7: Log In page 35

LIST OF TABLES

Table 1.1: Technologies Comparison of Similar E-Commerce Systems.....3
Table 1.2: Report Structure Overview.....7
Table 2.1: Functional Requirements.....8
Table 2.2: Use Case – User Login.....9
Table 2.3: Use Case – Add Product to Cart.....9
Table 2.4: Use Case – Place Order.....9
Table 2.5: Use Case – Generate Report.....9
Table 2.6: User Roles and Permission.....10
Table 2.7: Non-Functional Requirements.....10
Table 2.8: Hardware Requirements.....11
Table 2.9: Software Requirements.....11
Table 2.10: Database Scheme.....12
Table 2.11: Tools and Development Environment.....24
Table 2.12: System Modules and their description.....35
Table 2.13: Sample Test Cases for Online E-Commerce System.....27

LIST OF ABBREVIATIONS

Abbreviation	Full Form
BIM	Bachelor of Information Management
TPS	Transaction Processing System
MIS	Management Information System
UI	User Interface
SRS	System Requirements Specification
UAT	User Acceptance Testing
SME	Small and Medium Enterprise
API	Application Programming Interface
DBMS	Database Management System
HTML	Hypertext Markup Language
CSS	Cascading Style Sheet

CHAPTER I: INTRODUCTION

1.1 Background of the study

The rapid evolution of technology has reshaped consumer lifestyles and expectations, making digital connectivity a necessity in today's business landscape (Ms. Shipra & Mrs. Colin Janet, 2023). E-commerce has emerged as a powerful platform that simplifies the buying and selling of goods using the internet, offering consumers convenience, cost-effectiveness, and access to a global marketplace. Especially during the COVID-19 pandemic, when physical access to services was limited, digital platforms enabled uninterrupted transactions and contactless delivery. According to the study, E-commerce functions through various technologies like mobile commerce, online transaction processing, and digital marketing, offering a seamless customer experience. Popular platforms such as Amazon and Uber provide not only products but also customization and interaction, enhancing user engagement. However, with growing usage come security risks including data breaches and phishing, making consumer education and technological safeguards essential. Hence, understanding the dynamics of e-commerce is critical for ensuring secure, efficient, and customer-centric online transactions (IJSREM, 2023).

Electronic commerce (e-commerce) has transformed the way businesses conduct buying and selling activities by enabling transactions through digital platforms. With the rapid growth of internet usage and mobile technologies, customers now expect convenience, speed, and transparency when purchasing products. For businesses, especially small and medium-sized enterprises (SMEs), e-commerce systems provide an efficient way to reach customers, manage orders, and record sales transactions accurately.

In Nepal, many small retailers and local product manufacturers still rely on manual sales processes or basic social media pages for selling products. These traditional methods often lack proper order tracking, customer management, and transaction records, leading to inefficiencies, errors, and difficulties in monitoring business performance. Manual handling of sales and orders also makes it challenging to maintain data consistency and accountability.

To overcome these limitations, a web-based e-commerce system offers a structured and automated solution. Such a system records transactions electronically, manages customer information, and processes orders in real time. This project, E-Commerce System, is developed to automate online product sales by providing features such as user registration, product browsing, cart management, and order placement. The system aims to simplify transaction handling, reduce human errors, and provide a foundation for future analytical and managerial decision-making.

1.2 Problem Statement

Many small businesses face challenges in managing sales and customer transactions due to the absence of a centralized digital system. Traditional selling methods do not provide proper mechanisms for tracking customer orders, managing carts, or maintaining transaction history. As a result, businesses may experience data loss, inaccurate records, delayed order processing, and poor customer experience.

Furthermore, without automation, business owners find it difficult to analyze sales performance, monitor customer behavior, or generate reports for decision-making. The lack of a structured system also limits scalability, making it hard to expand operations as the number of customers and transactions increases. These issues highlight the need for a reliable, secure, and automated e-commerce system that can efficiently handle daily transactions and store accurate sales data.

1.3 Objectives of the Project

The main objectives of this project are as follows:

i. To develop a web-based e-commerce application for online product selling.
ii. To allow users to register, log in, browse products, and view product details.
iii. To implement cart functionality and enable users to place orders online.
iv. To record and process sales transactions accurately.
v. To provide a foundation for future reporting and management information features.

1.4 Literature Review

1.4.1 Overview of Existing Systems

Several well-known e-commerce platforms such as Amazon, Daraz, Shopify, and Flipkart provide complete online shopping solutions. These systems support product management, secure payments, order processing, and customer management. However, such platforms may be costly or overly complex for small local businesses. As a result, many SMEs prefer custom or lightweight e-commerce systems tailored to their specific needs.

1.4.2 Technologies Used in Similar Systems

Most modern e-commerce systems are developed using a combination of frontend, backend, database, and deployment technologies to ensure scalability, security, and efficient transaction processing.

Table 1.1: Technologies Comparison of Similar E-Commerce Systems

System / Project	Programming Language	Database	Framework	Notes
Amazon (Global E-	Java, Python, JavaScript	DynamoDB, Aurora, MySQL	Spring, React	Uses microservices architecture, cloud-native infrastructure

Commerce Platform)				(AWS), and advanced recommendation systems.
Shopify (E-Commerce SaaS)	Ruby, JavaScript	MySQL	Ruby on Rails, React	Provides scalable SaaS-based e-commerce solutions with secure payment and order processing.
Daraz (Regional E-Commerce Platform)	Java, JavaScript	MySQL	Spring Boot, Vue.js	Supports high-volume transactions, mobile commerce, and localized logistics integration.
Flipkart (Online Retail Platform)	Java, Python	MySQL, Cassandra	Spring, React	Uses distributed systems and data analytics for transaction handling and personalization.

1.4.3 Research and Academic Works

Transaction-based information systems and e-commerce platforms have been widely studied in the fields of information systems and financial technology. Researchers emphasize that Transaction Processing Systems (TPS) play a crucial role in recording, processing, and storing daily business transactions accurately and efficiently. Such systems form the operational backbone of organizations by ensuring data reliability and consistency (West, 2021).

Laudon and Laudon (2022) describe Transaction Processing Systems (TPS) as the foundation of operational-level information systems within organizations. They explain that TPS handles routine and repetitive transactions such as sales processing, order entry, billing, and payment recording with high speed and accuracy. By automating these core activities, TPS reduces manual workload and ensures reliable data capture, which is essential for daily business operations and long-term organizational stability.

West (2021) emphasizes that TPS plays a crucial role in ensuring data accuracy, consistency, and integrity across organizational processes. According to the study, reliable transaction data forms the base for higher-level systems such as Management Information Systems (MIS) and Decision Support Systems (DSS). Without accurate TPS data, analytical and managerial systems cannot function effectively, highlighting the strategic importance of transaction processing in modern information systems.

Turban et al. (2021) explain that modern e-commerce platforms rely heavily on TPS to automate customer interactions, order processing, and transaction recording in

real time. Their research highlights that TPS enables seamless coordination between customers, sellers, and system databases, ensuring faster service delivery and improved user experience. This automation is critical for handling high volumes of online transactions efficiently.

Campbell, Martinez-Jerez, and Tufano (2022) found that automated transaction systems significantly reduce human errors and improve compliance when compared to manual processes. Their research shows that system-generated records minimize inconsistencies and delays while enforcing standardized procedures. This improves operational accuracy and enhances trust in digital transaction systems.

Wang (2021) highlights that centralized digital transaction systems enhance transparency and traceability within organizations. By maintaining structured and centralized records, organizations can monitor transactions more effectively, detect anomalies, and improve internal control mechanisms. This level of transparency is particularly important for audit readiness and regulatory compliance.

Bakoben, Giudici, and Figini (2020) demonstrate that structured transactional data collected through digital systems improves prediction accuracy and operational planning. Their study emphasizes that reliable transaction records allow organizations to analyze patterns and trends, leading to better forecasting and informed decision-making. Such benefits are only possible when TPS is properly implemented.

O'Brien and Marakas (2020) state that web-based information systems improve accessibility and operational efficiency, especially for small and medium-sized enterprises (SMEs). Their work highlights that online systems allow businesses to manage operations remotely while reducing infrastructure costs. This makes web-based TPS solutions highly suitable for SMEs with limited resources.

Chaffey and Ellis-Chadwick (2019) argue that e-commerce systems expand market reach by enabling businesses to operate beyond geographical boundaries. Their research explains that digital platforms allow organizations to attract a wider customer base while operating continuously. TPS supports this expansion by ensuring reliable transaction handling regardless of time and location.

Kumar and Reinartz (2018) emphasize the importance of customer data management in digital platforms to enhance personalization and customer engagement. They explain that accurate transaction histories allow businesses to understand customer preferences and behavior. TPS plays a vital role in capturing and maintaining this customer-related data.

Heizer, Render, and Munson (2020) explain that automation in order processing improves processing speed, accuracy, and customer satisfaction. Their research

indicates that automated systems reduce delays and errors associated with manual order handling. This results in faster delivery cycles and improved service quality.

Le et al. (2023) identify system scalability and innovation capacity as key drivers of sustainable digital platform growth. Their study highlights that transaction systems must be capable of handling increasing data volumes as business operations expand. Scalable TPS architecture ensures long-term system usability and performance.

Bhatt, Chennells, and Vause (2023) highlight that digital platforms require robust transaction handling mechanisms to maintain financial stability and user trust. They argue that system failures or transaction inaccuracies can significantly damage customer confidence. Reliable TPS design is therefore essential for sustaining digital business operations.

Laudon (2020) states that reliable transaction systems improve organizational efficiency by reducing operational costs and processing time. Automation minimizes redundant tasks and optimizes workflow efficiency, allowing organizations to allocate resources more effectively.

Satzger, Psaier, and Hummer (2021) discuss how modular system design enables e-commerce platforms to adapt to future technological changes. They explain that modular TPS architecture allows new features to be integrated without disrupting existing operations, supporting long-term system evolution.

Peffers et al. (2018) emphasize that design-oriented information system projects should focus on solving real business problems through functional system development. Their research supports the development of practical systems like e-commerce TPS that directly address operational challenges.

Jakubik and Teleu (2025) explain that historical transaction data stored in TPS supports future predictive and analytical models. Accurate transaction records serve as a foundation for advanced analytics, forecasting, and management information systems.

Zwass (2019) defines e-commerce systems as transaction-centric platforms that integrate customers, businesses, and digital processes. The study highlights that TPS acts as the core mechanism enabling secure and efficient interactions among all stakeholders.

Dennis, Wixom, and Roth (2021) highlight that structured system development methodologies improve system reliability and maintainability. Proper design and documentation ensure that transaction systems remain stable and adaptable over time.

Chen, Chiang, and Storey (2019) argue that high-quality transactional data is essential for future analytics and business intelligence applications. TPS ensures data quality by capturing accurate and consistent transaction records at the operational level.

Alter (2022) concludes that well-designed transaction processing systems provide long-term organizational value by supporting both operational efficiency and strategic decision-making. TPS serves as a critical link between day-to-day operations and higher-level management systems.

Overall, the reviewed literature supports the development of e-commerce and financial systems as Transaction Processing Systems, which can later be enhanced into Management Information Systems (MIS) by incorporating reporting, analytics, and decision-support features.

1.5 Development Methodology

The project follows the Iterative Waterfall Model as its development methodology. This model combines the structured phases of the traditional waterfall approach requirements analysis, system design, implementation, testing, and deployment with iterative feedback and refinement. Each phase is completed systematically, while allowing improvements based on testing and user feedback.

This methodology ensures that the e-commerce system is developed in a controlled and organized manner. The iterative nature allows issues identified during development to be corrected early, resulting in a stable and reliable application. The approach is suitable for this project as it ensures clarity in requirements and gradual enhancement of features.

Figure 1.1: Iterative Waterfall Model for E-Commerce System

1.6 Scope and Limitation of the Project

The E-Commerce System is designed to support online selling by providing a centralized platform for managing products, users, carts, and orders. It enables customers to browse products, add items to a cart, and place orders online. The system records all transactions electronically, making it suitable as a Transaction Processing System (TPS). The project also provides scope for future enhancements such as sales reports, analytics dashboards, and inventory tracking, which can extend the system into a Management Information System (MIS).

The system is particularly suitable for small and medium-sized enterprises, local sellers, and startup businesses seeking a simple yet effective online sales platform.

Limitations

Some limitations of the E-Commerce System include:

i. The system requires a stable internet connection as it is web-based.
ii. Online payment gateway integration may be limited or simulated.
iii. Advanced analytics and reporting features are not included in the current version.
iv. Inventory management and supplier modules are not fully implemented.

1.7 Report Organization

This section of the report provides an overview of how the BIM Minor Project Report is structured.

Table 1.2: Report Structure Overview

Chapter No.	Chapter Title	Description
Chapter I	**Introduction**	Provides background, problem statement, objectives, related works, methodology, scope, and report organization.
Chapter II	**System Development Process**	Discusses system analysis, design, requirements, feasibility, and system architecture.
Chapter III	**Conclusion and Recommendations**	Explains tools, frameworks, and the development process with coding and database design.
References		Lists all sources, journals, and materials referred to.
Appendices		Includes screenshots, code snippets, and supporting documents

CHAPTER II: SYSTEM DEVELOPMENT PROCESS

2.1 Analysis

The analysis phase of the E-Commerce System focused on identifying limitations in traditional selling and order management practices and determining how these processes could be automated through a web-based platform. Many small and medium-sized businesses rely on manual sales tracking, social media messaging, or informal records, which often leads to inaccurate order data, poor customer management, and difficulty in monitoring transactions.

To address these challenges, requirement analysis was conducted by studying existing e-commerce platforms, analyzing user expectations, and evaluating transaction-handling practices. The analysis revealed the need for a system that can manage users, products, carts, and orders in a structured and automated manner. The findings were translated into functional and non-functional requirements to guide the design and implementation of the system.

2.1.1 Requirement Analysis

Requirement analysis was performed to bridge the gap between traditional selling methods and the proposed automated e-commerce solution. The core requirements included secure user authentication, product browsing, cart management, order placement, and transaction recording. Usability, performance, and security were also identified as critical system requirements.

2.1.1.1 Functional Requirements

The system is expected to provide the following functionalities:

Table 2.1: Functional Requirements

Requirement ID	Requirement Description	Example / Details
FR1	User Authentication & Authorization	Users must register and log in to access system features.
FR2	Product Management	Display products with name, price, description, and images.
FR3	Cart Management	Users can add, update, or remove items from the cart.
FR4	Order Processing	Place orders and store order details securely.
FR5	Transaction Recording	Record all sales transactions accurately.
FR6	Order History & Status	Users can view past orders and order status.
FR7	Admin Dashboard	Admin can manage products and view all orders.

2.1.1.2 Use Case Scenario

A case scenario describes the interaction between users and the system to achieve a specific goal. The use cases of the system are discussed in the tables below:

Table 2.2: Use Case – User Login

Use Case	**User Login**
Actors	User, Admin
Preconditions	User must be registered
Steps	1. Enter email and password 2. Click Login 3. System validates credentials 4. Redirect to dashboard
Postcondition	User logged in or error message shown

Table 2.3: Use Case – Add Product to Cart

Use Case	**Add to Cart**
Actors	User
Preconditions	User must be logged in
Steps	1. Select products 2. Choose quantity 3. Click "Add to Cart"
Postcondition	Product added to cart

Table 2.4: Use Case – Place Order

Use Case	**Place Order**
Actors	User
Preconditions	Cart contains products
Steps	1. Review Cart 2. Enter delivery details 3. Confirm order
Postcondition	Order successfully placed

Table 2.5: Use Case – Generate Report

Use Case	**Generate Sales Report**
Actors	Admin
Preconditions	Admin logged in
Steps	1. Select report type 2. System generates report
Postcondition	Sales report displayed

2.1.1.3 User Roles & Permissions

The system defines different types of users with specific roles and access levels to ensure security and efficient workflow:

Table 2.6: User Roles and Permission

Role	Permissions
Admin	Manage products, view all orders, generate reports
User	Browse products, manage carts, place orders
System	Processes transactions and updates order records

2.1.1.4 Non-Functional Requirements

Non-functional requirements are the specifications that describe the system's operation capabilities, constraints, and how it should operate, focusing on quality attributes like performance, usability, compatibility, and many more. These requirements define how the system should be performed. The non-functional requirements for online order management system are given below:

Table 2.7: Non-Functional Requirements Table

Requirement ID	Requirement	Description
NFR1	Usability	Simple and intuitive interface
NFR2	Security	Secure login and protected data
NFR3	Performance	Fast page load and transaction processing
NFR4	Scalability	Supports increasing users and orders
NFR5	Compatibility	Works on modern browsers and mobile devices
NFR6	Maintainability	Modular and easy to update

2.1.2 Feasibility Study

i. **Technical Feasibility:** Uses web technologies such as HTML, CSS, JavaScript, and backend frameworks suitable for e-commerce.
ii. **Economic Feasibility:** Open-source tools minimize development costs.
iii. **Operational Feasibility:** Easy-to-use interface suitable for small businesses.
iv. **Legal Feasibility:** Complies with basic data protection standards.
v. **Schedule Feasibility:** Achievable timeline using Iterative Waterfall Model.

2.1.3 System Requirements Specification (SRS)

Below are minimum and ideal hardware and software requirements for optimal performance, scalability, and security.

Table 2.8: Hardware Requirements

Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5
RAM	4 GB	8 GB
Storage	128 GB	256 GB
Internet	5 Mbps	10 Mbps

Table 2.9: Software Requirements

Component	Specification
Operating System	Windows / Linux
Frontend	HTML, CSS, JavaScript
Backend	Django
Database	SQLite (development), PostgreSQL (deployment)
Browser	Chrome, Firefox, Edge

2.2 System Design

System design is a crucial phase in the development of the E-Commerce System, as it translates system requirements into a structured blueprint for implementation. This phase defines the overall architecture, database structure, user interface layout, system workflow, and object-oriented models. The goal of system design is to ensure that the application is efficient, scalable, secure, and easy to use while supporting transaction processing accurately.

The design of the e-commerce system focuses on handling daily sales transactions, managing users and products, and storing order-related data in a centralized database. The system is designed to function primarily as a Transaction Processing System (TPS), with the ability to expand into a Management Information System (MIS) in the future.

2.2.1 System Architecture

The E-Commerce System is designed using a three-tier client–server architecture, which separates the application into presentation, application logic, and data layers. This architecture improves maintainability, scalability, and security. The system consists of:

i. **Frontend:** User interface for browsing and ordering products
ii. **Backend:** Handles business logic and transactions
iii. **Database:** Stores user, product, and order data

This layered design ensures smooth communication between users and the system while maintaining data integrity during transaction processing.

2.2.2 Database Design

Database design defines how data is structured, stored, and accessed within the E-Commerce System. A well-designed database ensures data consistency, reduces redundancy, and supports efficient transaction processing. Since the system handles frequent order-related transactions, the database plays a critical role in ensuring reliability and accuracy.

The database stores information related to users, products, carts, orders, and transactions. Relationships between entities are designed to support fast retrieval of order history, user activity, and product details.

2.2.2.1 Database Schema

A database schema represents the logical structure of the database, including tables, attributes, and relationships. It acts as a blueprint for organizing data in the system.

Table 2.10: Database Scheme

Table Name	Attributes	Description	Relationship
Users	user_id, name, email, password, role	Stores registered user and admin information	Primary table
Products	product_id, name, price, description, stock	Stores product details available for sale	Referenced by Cart and Orders
Cart	cart_id, user_id, product_id, quantity	Temporarily selected items	Linked with Users and Products
Orders	order_id, user_id, order_date, total_amount, status	Stores order information	Linked with Users
Order_Items	item_id, order_id, product_id, quantity, price	Stores individual items of each order	Linked with Orders and Products
Transactions	transaction_id, order_id, payment_mode, date	Stores transaction records	Linked with Orders

This database structure enables the system to efficiently record and retrieve transaction data, making it suitable for TPS operations.

The Entity Relationship Diagram (ERD) represents the logical structure of the Online E-Commerce System database. It illustrates the relationships between core entities such as User, Product, Cart, Order, Order_Item, and Transaction. The ERD ensures data consistency and supports efficient transaction processing by clearly defining primary keys, foreign keys, and entity relationships. This structure enables accurate recording of sales transactions, order history, and customer activities, making the system suitable for Transaction Processing System (TPS) operations.

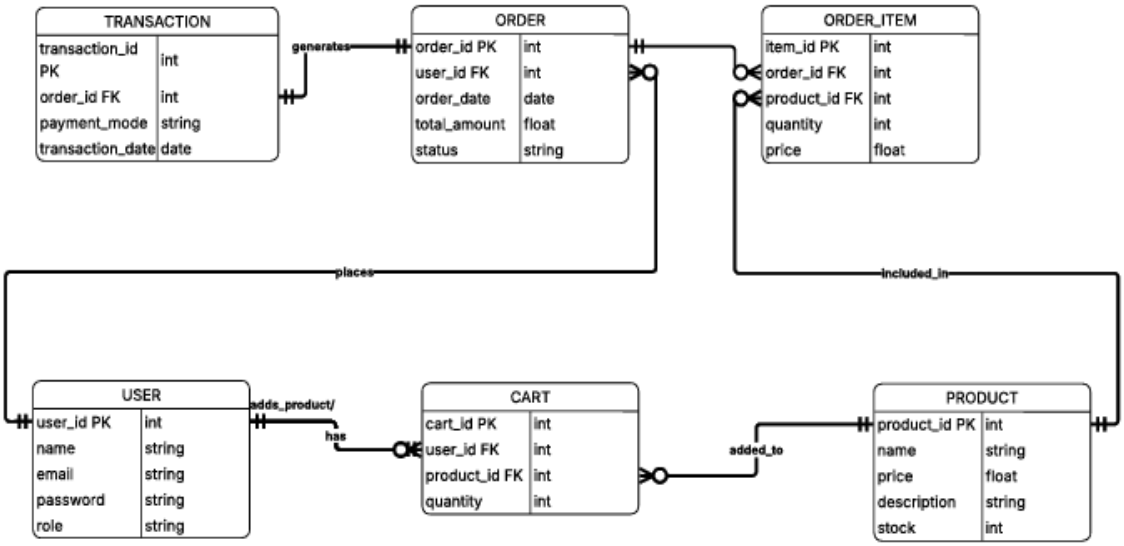

Figure 2.1: Entity Relationship Diagram

2.2.3 User Interface Design

The User Interface (UI) of the E-Commerce System is designed to provide simple, intuitive, and responsive experience for users and administrators. The design emphasizes ease of navigation, clarity of information, and minimal user effort to complete transactions.

The UI is designed following usability principles so that even non-technical users can browse products, manage carts, and place orders without difficulty.

2.2.3.1 Major UI Components

Dashboard: The dashboard provides a summary view of system activity.

i. Admin Dashboard: Displays total products, total orders, and sales summary
ii. User Dashboard: Displays recent orders and account information

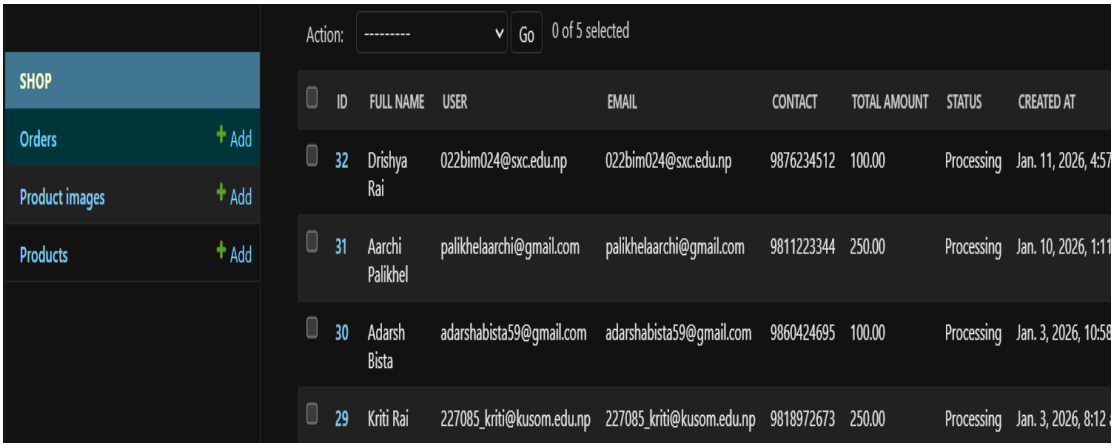

Figure 2.2: Admin Dashboard

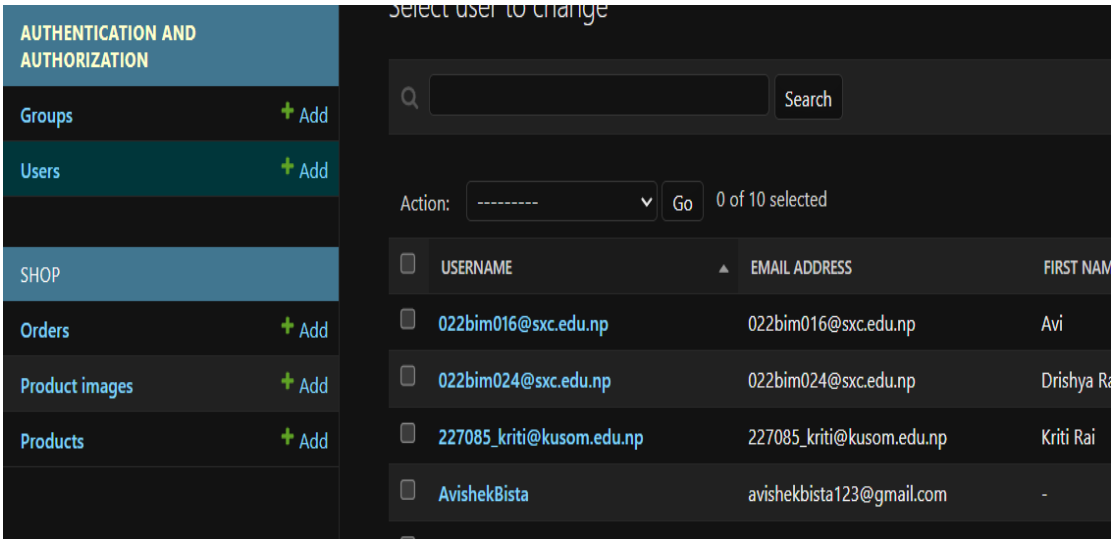

Figure 2.3: User Dashboard

Pages in the System

i. Login & Signup Page: Allows users to register and log in securely

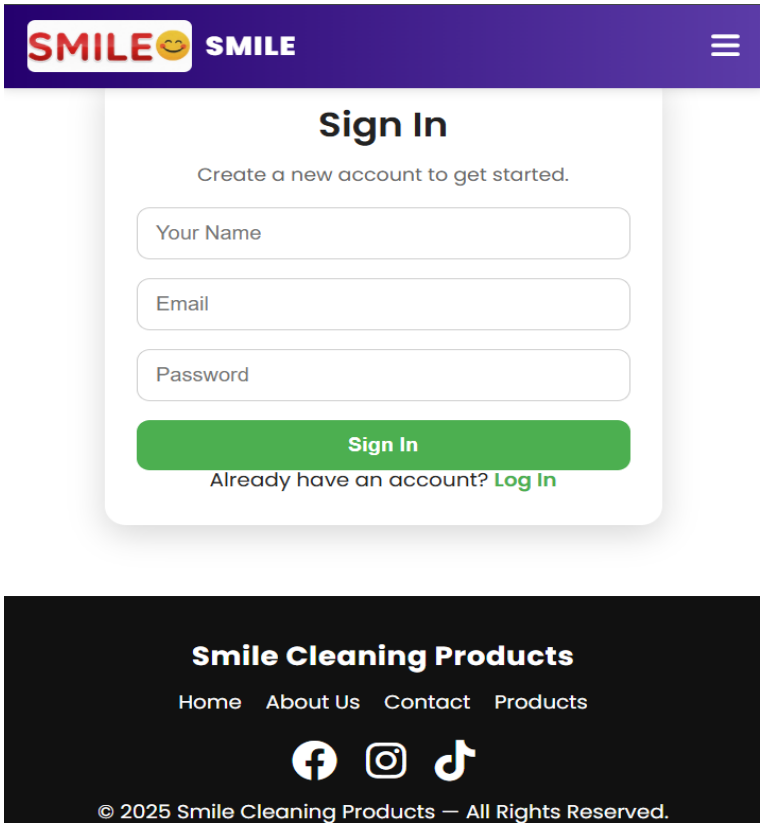

Figure 2.4: Sign In Page

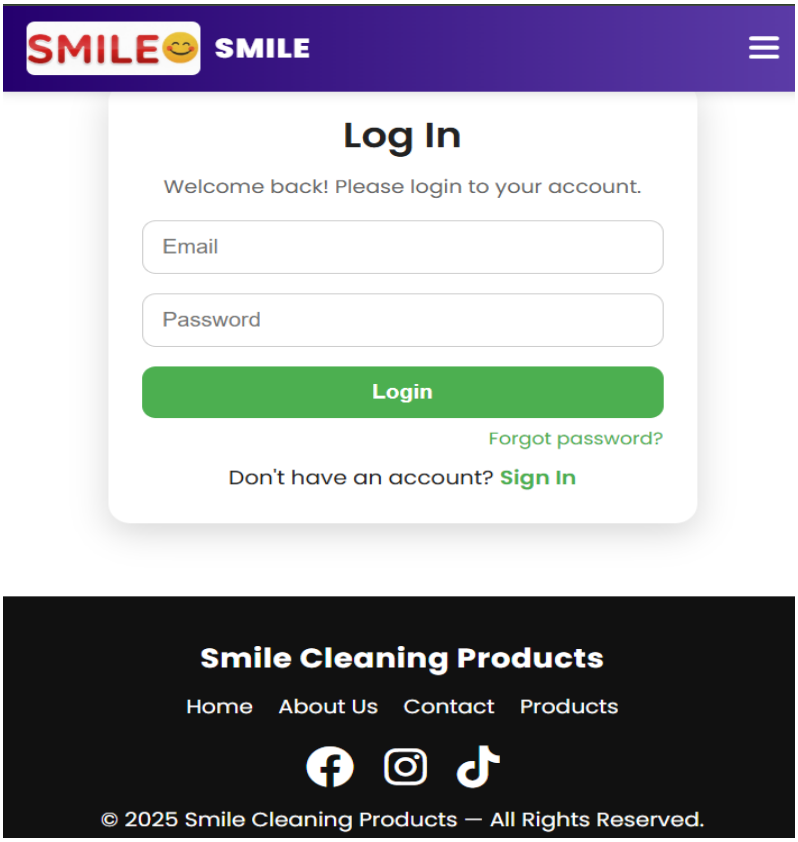

Figure 2.5: Log In Page

ii. Home Page: Displays featured and available products

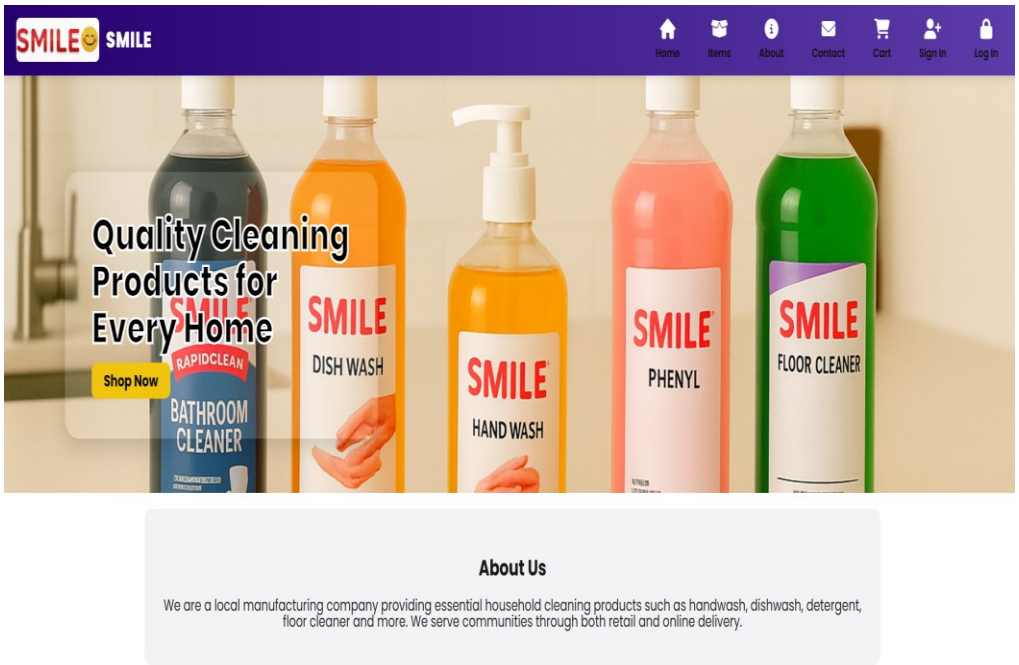

Figure 2.6: Home Page

iii. Product Listing Page: Shows all products with price and description

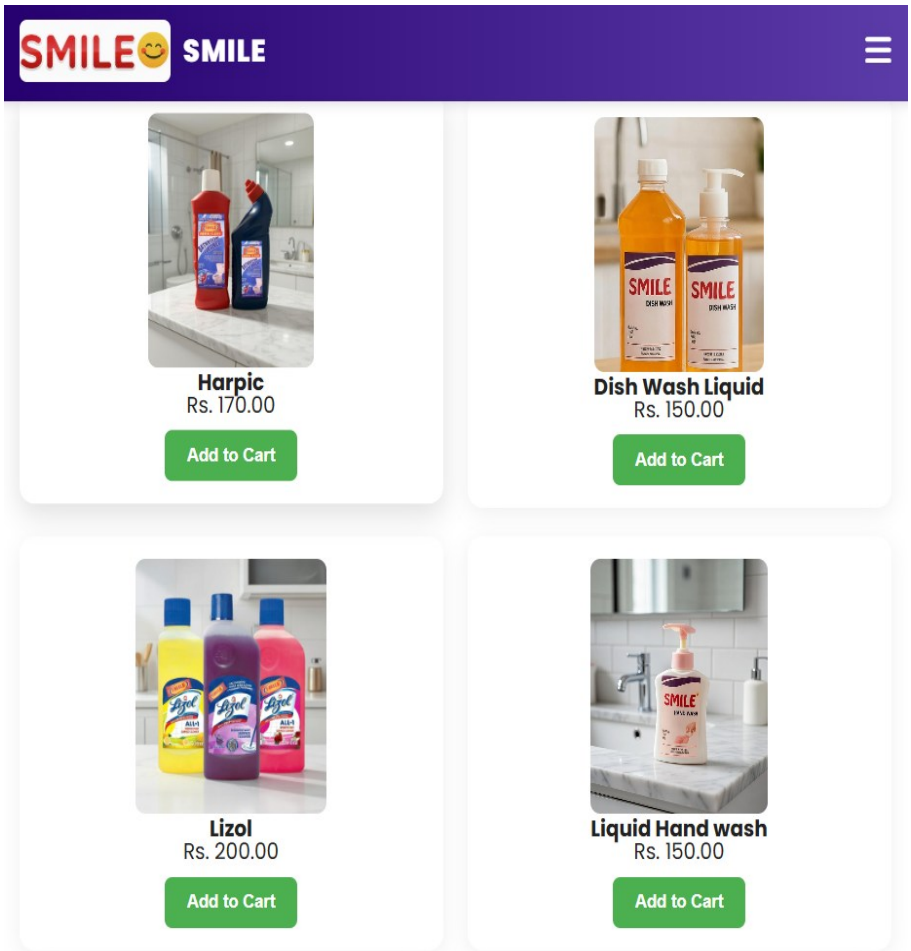

Figure 2.7: Product listing Page

iv. Product Detail Page: Displays detailed information about a selected product

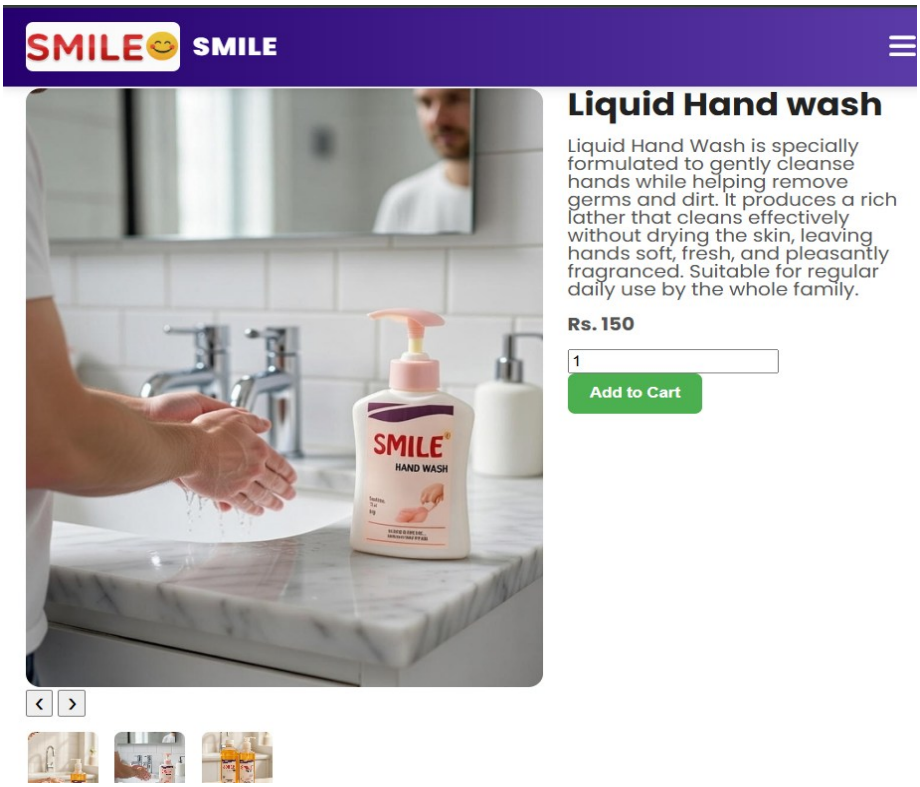

Figure 2.8: Product Detail Page

v. Cart Page: Shows selected items and total cost

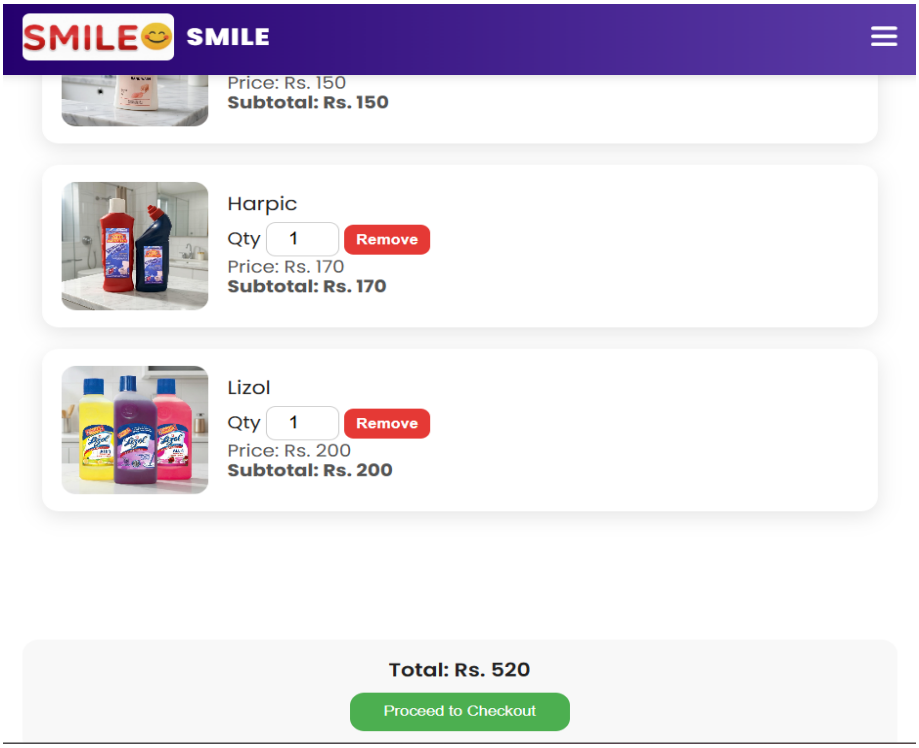

Figure 2.9: Cart Page

vi. Checkout Page: Allows order confirmation

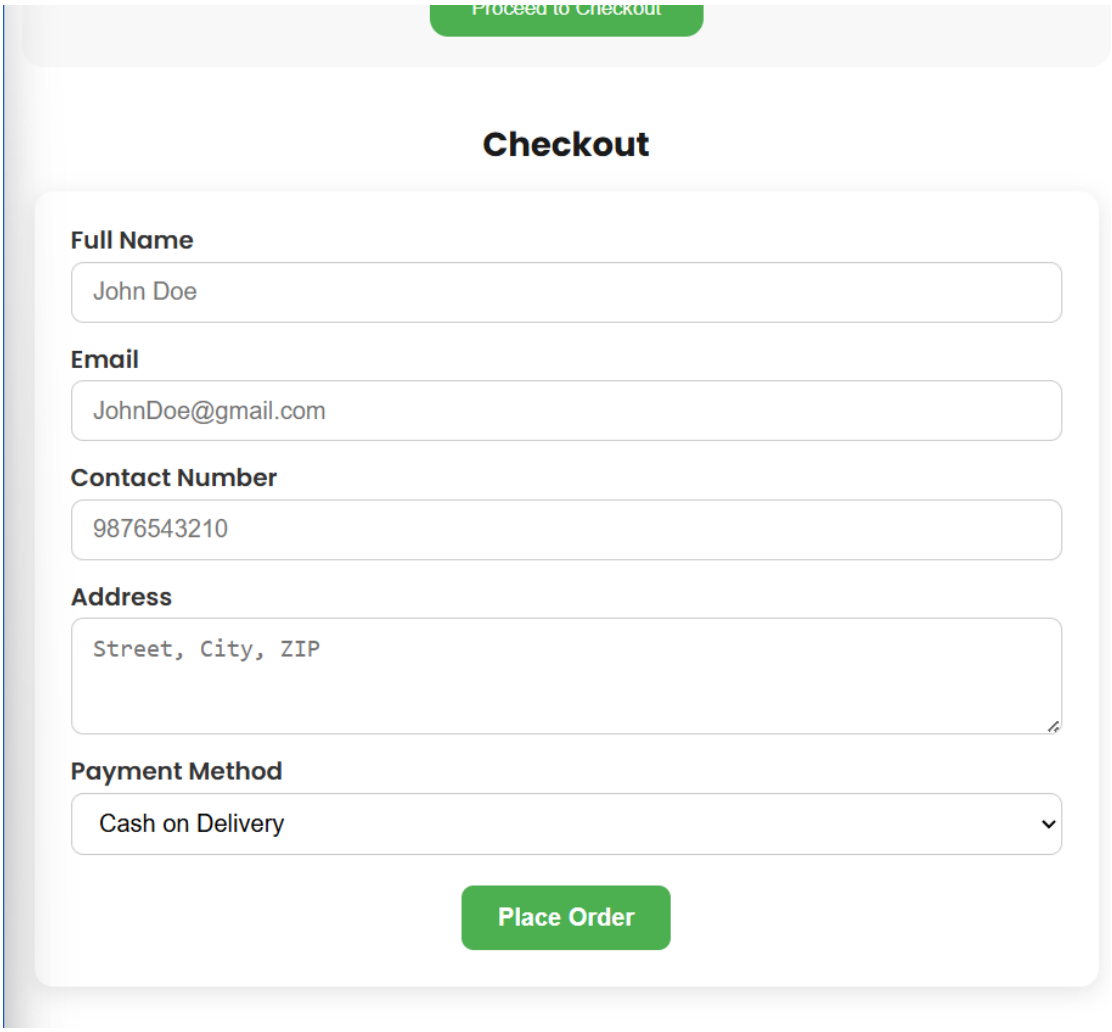

Figure 2.10: Checkout Page

2.2.4 System Flowchart

A system flowchart represents the logical flow of activities within the E-Commerce System. It illustrates the sequence of steps involved in completing a transaction, from user login to order placement.

The flowchart includes:

- User authentication
- Product selection
- Cart management
- Order confirmation
- Transaction recording
- Order completion

The flowchart helps visualize how data moves through the system and how decisions are made at each stage.

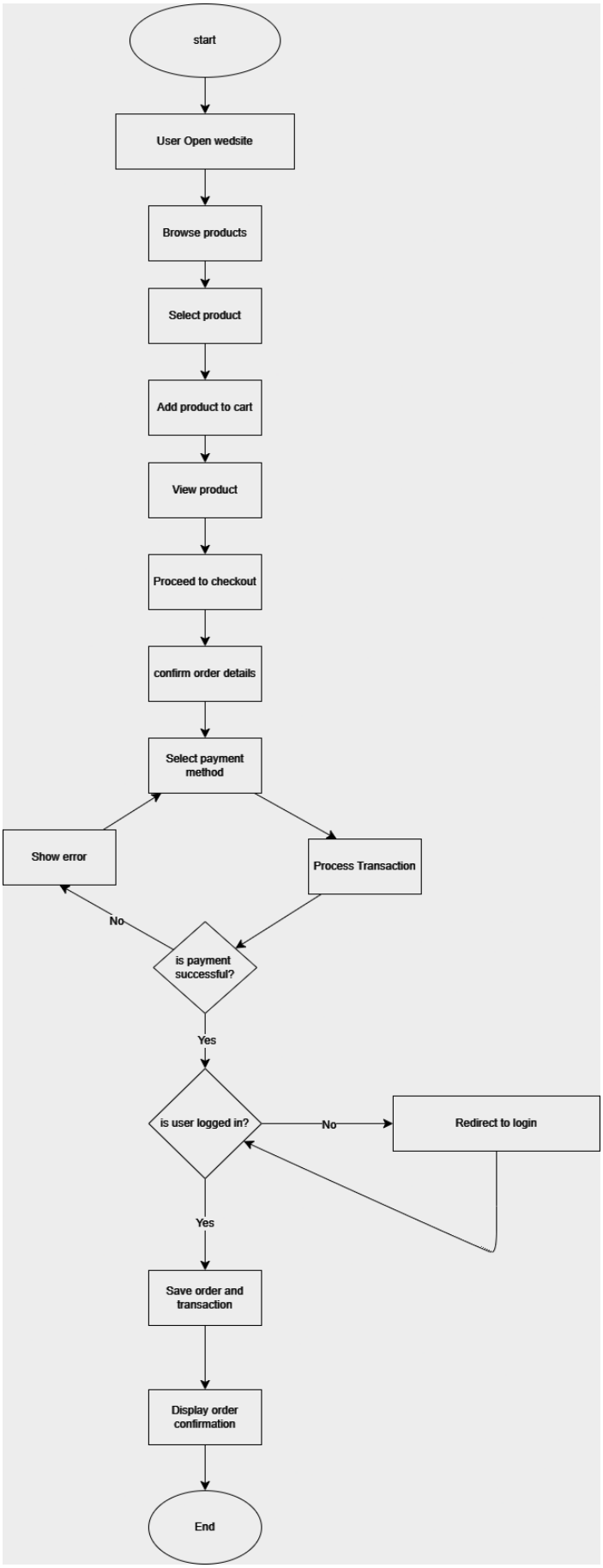

Figure 2.11: Flowchart of Online E-Commerce System

2.2.5 Object-Oriented Modelling

Object-oriented modeling is used to represent the system using objects, classes, and relationships. This approach improves modularity, reusability, and maintainability of the system.

2.2.5.1 System Overview

The E-Commerce System is a web-based application designed to manage online buying and selling activities. It enables users to browse products, place orders, and track transactions, while administrators manage products and monitor sales.

The system uses object-oriented principles to organize functionalities into reusable components such as User, Product, Cart, Order, and Transaction classes. This design supports scalability and easy feature enhancement.

2.2.5.2 Use Case Model

A case model provides a high-level view of system functionality by showing interactions between users and the system.

Primary Actors:

- User
- Admin

Major Use Cases:

- Register / Login
- Browse Products
- Add to Cart
- Place Order
- View Order History
- Manage Products (Admin)
- Generate Reports (Admin)

The case model ensures that all functional requirements identified during analysis are properly represented and implemented in the system.

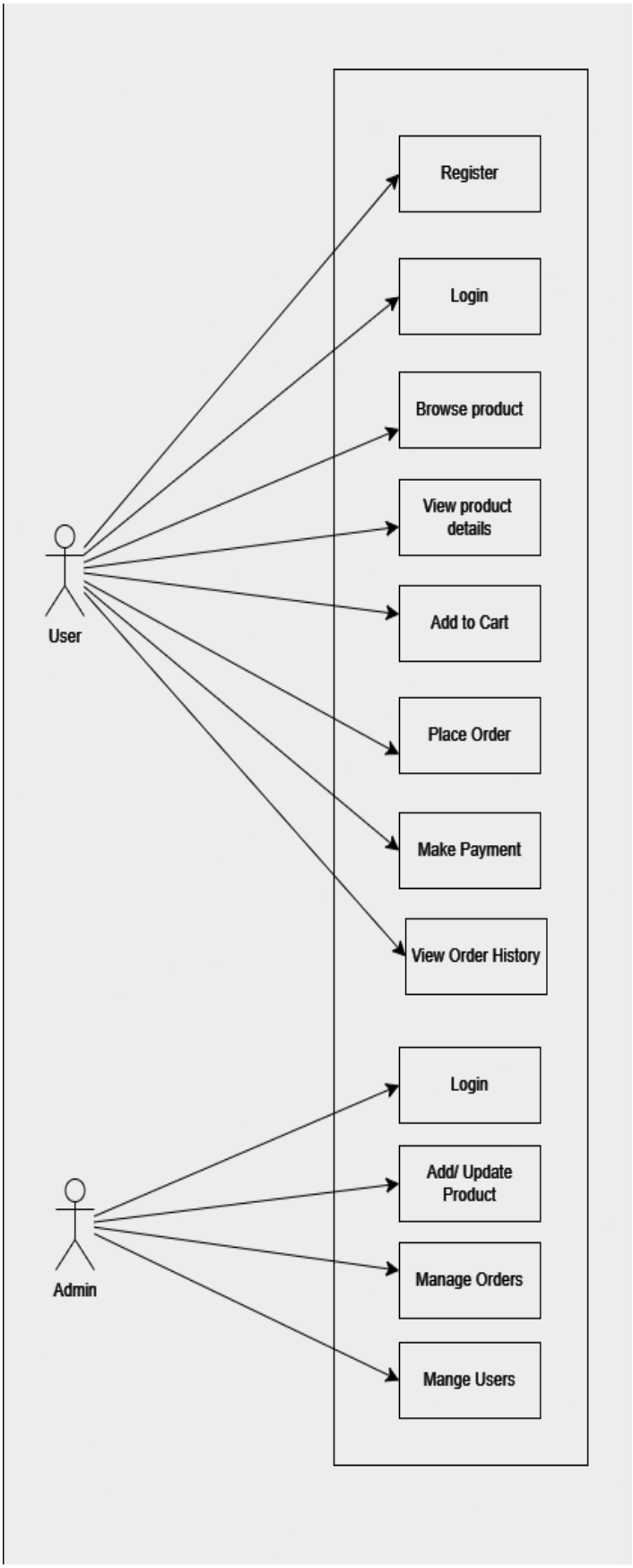

Figure 2.12: Use Case Diagram of Online E-Commerce System

2.2.5.3 Class Diagram

The Class Diagram represents the static structure of the Online E-Commerce System using object-oriented principles. It defines the system classes, their attributes, methods, and relationships. The diagram demonstrates how users interact with products through carts and orders, while transactions record completed purchases. This modular design improves system maintainability, reusability, and scalability.

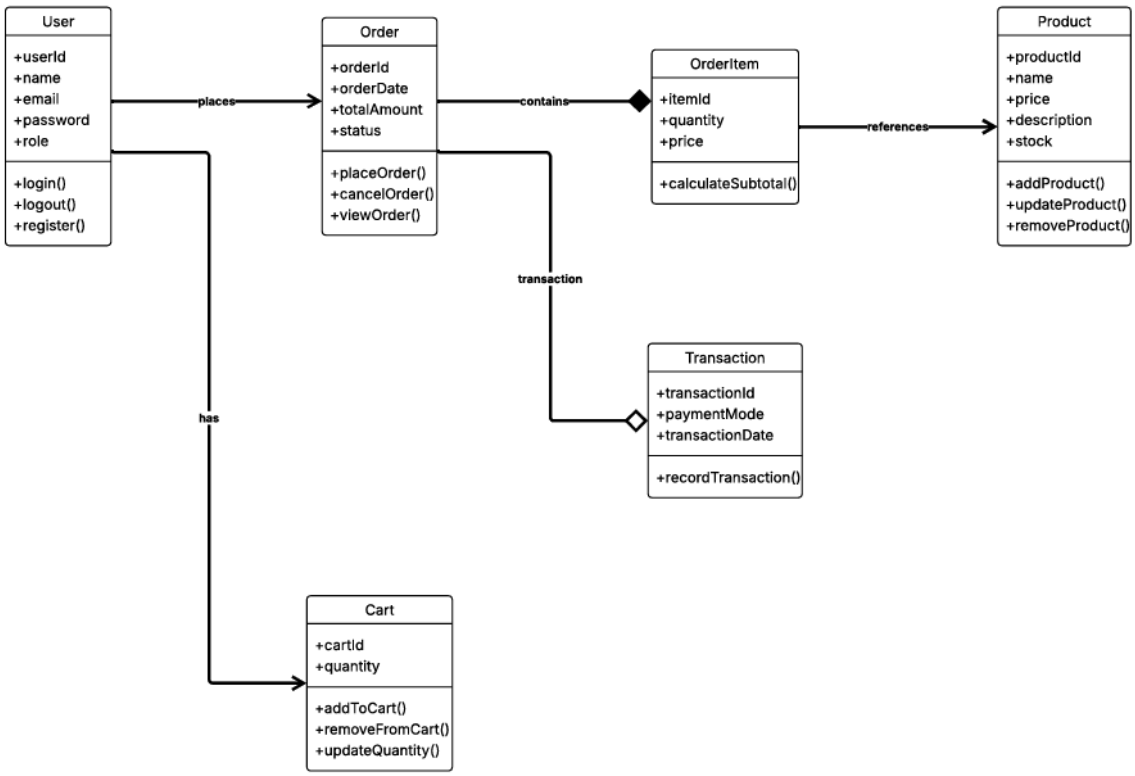

Figure 2.13: Class Diagram of Online E-Commerce System

2.2.5.4 Object Diagram

The Object Diagram shows a runtime instance of the Online E-Commerce System. It represents real objects created from classes during system execution. This diagram illustrates how a user selects a product, adds it to a cart, places an order, and completes a transaction. It provides a practical view of system behavior at a specific point in time.

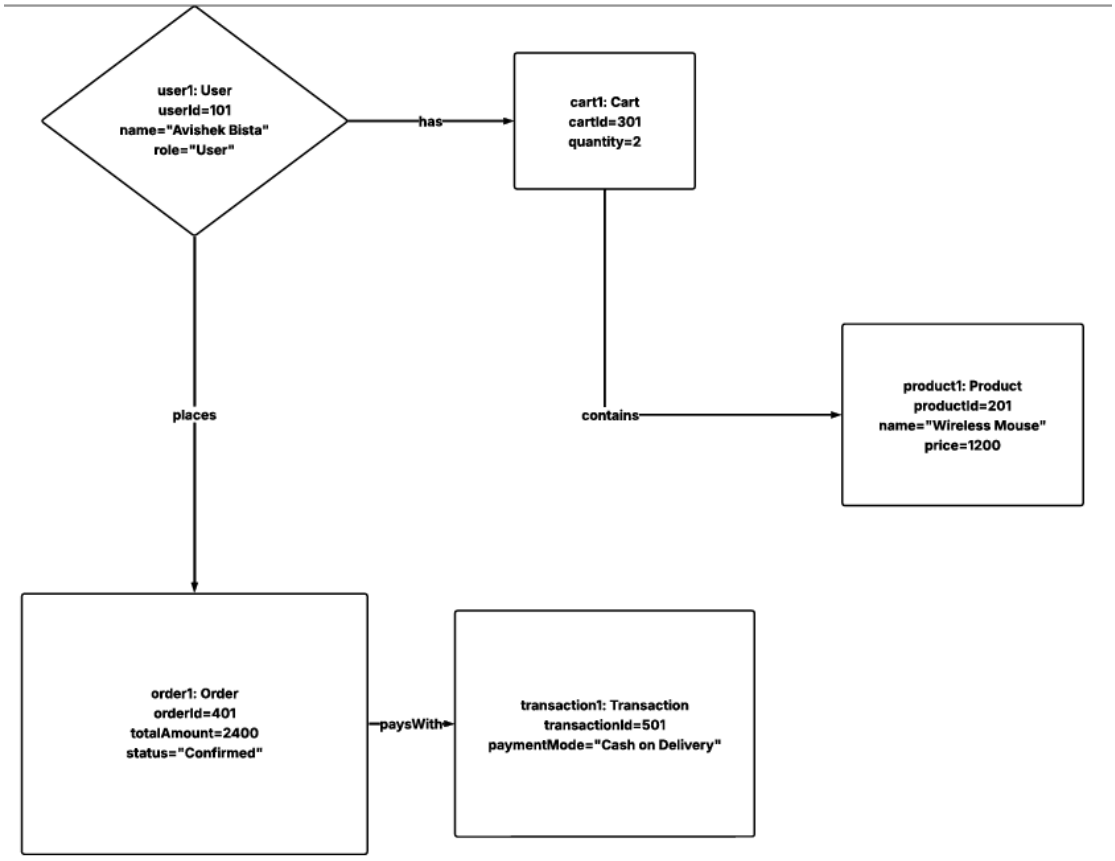

Figure 2.14: Object Diagram of Online E-Commerce System

2.3 Implementation

The system was implemented as a web-based application integrating frontend and backend through RESTful APIs. The implementation focused on secure authentication, accurate order processing, and smooth user interaction.

2.3.1 Tools and Development Environment

Table 2.11: Tools and Development Environment

Tool	Category	Purpose
HTML/CSS	Frontend	Page structure and styling
JavaScript	Logic	Dynamic interaction
Django (Python)	Backend	Server-side processing
SQLite (development)	Database	Data storage
VS Code	IDE	Development

2.3.2 System Modules and their Implementation

The E-Commerce System is divided into several functional modules that work together to support online buying and selling activities. Each module is designed as an independent yet interconnected component, ensuring scalability, maintainability, and efficient transaction processing. The modular design allows the system to function

effectively as a Transaction Processing System (TPS) while supporting future enhancement into a Management Information System (MIS).

Table 2.12: System Modules and their description

Module	Description
User Authentication Module	Manages user registration, login, logout, and session handling. Users must be authenticated before placing orders. The system differentiates between users and administrators, granting role-based access to dashboards and functionalities.
Product Management Module	Allows administrators to add, update, and remove products. Product details such as name, price, description, stock quantity, and images are stored and displayed dynamically to users.
Cart Management Module	Enables users to add products to a shopping cart, update quantities, and remove items. The cart data is maintained per user session to ensure continuity until checkout.
Order Management Module	Handles order placement, order confirmation, and order status tracking. Each order is linked to a specific user and stored in the database for transaction history.
Transaction Module	Records all purchase transactions, including payment mode, order amount, and transaction date. Each transaction is time-stamped to support accurate transaction processing and reporting.
Admin Management Module	Provides administrators with tools to manage users, products, and orders. Admins can monitor system activity and ensure smooth business operations.
Dashboard Module	Displays summarized information such as total products, total orders, and recent transactions. The admin dashboard supports monitoring system performance, while users can view order history and account details.

Each module interacts with the centralized database to ensure accurate data storage and retrieval, supporting real-time transaction processing.

2.3.3 Coding Standards and Conventions

To maintain code quality, readability, and scalability, standardized coding practices were followed throughout the development of the E-Commerce System.

i. Naming Conventions

- Variables and functions follow **camelCase** (e.g., addToCart(), placeOrder()).
- Class and model names follow **PascalCase** (e.g., User, Product, Order).
- File names are written in lowercase or PascalCase depending on usage (e.g., main.js, ProductPage.html).

ii. File Organization

The project follows a modular structure separating frontend and backend logic:

```
/ecommerce_project
│── manage.py
│
│── /ecommerce_project
│   │── settings.py
│   │── urls.py
│   │── wsgi.py
│
│── /accounts
│   │── models.py
│   │── views.py
│   │── urls.py
│   │── templates/
│
│── /products
│   │── models.py
│   │── views.py
│   │── urls.py
│
│── /orders
│   │── models.py
│   │── views.py
│   │── urls.py
│
│── /static
│── /templates
```

This structure improves maintainability and allows independent development of frontend and backend components.

iii. Documentation

- Inline comments were added to explain complex logic.
- Functions and modules were named descriptively to reflect their purpose.
- Code structure was kept consistent to support future development and debugging.

iv. Best Practices

- Consistent indentation and formatting were maintained.
- Reusable components and functions were used to reduce redundancy.
- Input validation was applied on both frontend and backend.

- Sensitive configuration data was isolated from core logic.

2.3.4 Security Considerations

Security was carefully considered during the design and development of the E-Commerce System to protect user data and transactional information.

i. **User Authentication Security:** User login credentials are securely handled, and only authenticated users are allowed to place orders or access sensitive features.
ii. **Input Validation:** All user inputs, such as login credentials, order quantities, and form data, are validated to prevent invalid or malicious entries.
iii. **Access Control:** Administrative functions such as product management and order monitoring are restricted to authorized users only.
iv. **Secure Deployment (Future Scope):** For future deployment, the system will use HTTPS to ensure encrypted communication between the client and server.

These measures ensure confidentiality, integrity, and availability of system data.

2.3.5 Testing and Debugging

Testing was conducted at multiple levels to ensure the accuracy, reliability, and stability of the E-Commerce System.

2.3.5.1 Unit Testing

Individual modules such as login validation, cart updates, and order placement were tested independently to ensure correct functionality.

2.3.5.2 Integration Testing

Integration testing verified smooth interaction between modules, such as cart data being correctly transferred to the order and transaction modules during checkout.

2.3.5.3 System Testing

The complete workflow from user login to order confirmation was tested to ensure the system functions as intended.

2.3.5.4 User Acceptance Testing (UAT)

Users and administrators tested the system under realistic conditions. Feedback was used to improve usability and performance.

Table 2.13: Sample Test Cases for Online E-Commerce System for Sales and Order Management

Test ID	Scenario	Steps	Expected Output	Status
TC_AUTH_01	Valid Login	Enter correct credentials	Redirect to dashboard	Pass

TC_AUTH_02	Invalid Login	Enter wrong credentials	Error message displayed	Pass
TC_CART_01	Add to Cart	Select product and add	Product added to cart	Pass
TC_ORD_01	Place Order	Confirm checkout	Order successfully placed	Pass
TC_TRANS_01	Record Transaction	Complete payment	Transaction saved	Pass
TC_ADM_01	Add Product	Admin submits form	Product listed	Pass

2.4 Deployment

The E-Commerce System was developed and tested in a local environment before planning for online deployment.

2.4.1 Local Deployment Configuration

i. **Frontend:** HTML, CSS, JavaScript rendered via Django templates
ii. **Backend:** Django development server (python manage.py runserver)
iii. **Database:** SQLite

2.4.2 Planned Online Deployment

i. **Backend:** Deployed on platforms such as PythonAnywhere, Railway, or AWS EC2
ii. **Database:** PostgreSQL (recommended)
iii. **Static Files:** Served via cloud storage
iv. **Security:** HTTPS enabled using SSL certificates

CHAPTER III: CONCLUSION AND RECOMMENDATIONS

3.1 Summary

This project titled "Online E-Commerce System for Sales and Order Management" was developed to address the challenges faced by small and medium-sized businesses in managing sales, orders, and transaction records using traditional or manual methods. Many local sellers in Nepal still rely on informal processes such as handwritten records or social media messaging, which often result in data inconsistency, poor order tracking, and limited business insights.

The proposed system was designed and implemented as a web-based e-commerce application that automates essential business functions including user registration and authentication, product browsing, cart management, order placement, and transaction recording. The system follows the Iterative Waterfall Development Model, ensuring a structured yet flexible approach to development.

The application was built using modern web technologies, with a clear separation of frontend, backend, and database layers. The system primarily functions as a Transaction Processing System (TPS) by recording daily sales and order transactions accurately and securely. It also provides a solid foundation for future enhancement into a Management Information System (MIS) by enabling structured data storage suitable for reporting and analytics.

Testing at unit, integration, system, and user acceptance levels confirmed that the system meets its functional and non-functional requirements. Overall, the project successfully demonstrates how an automated e-commerce system can improve efficiency, accuracy, and transparency in sales and order management.

3.2 Conclusion

The Online E-Commerce System for Sales and Order Management fulfills its primary objective of providing a reliable, structured, and user-friendly platform for managing online sales transactions. By replacing manual and semi-digital practices with an integrated web-based system, the project addresses key problems such as inaccurate record-keeping, delayed order processing, and lack of transaction history.

The system effectively supports essential e-commerce operations including secure user authentication, product management, cart handling, order processing, and transaction recording. Its modular architecture ensures maintainability and scalability, making it suitable for small and medium-sized enterprises seeking a cost-effective digital solution.

From an academic and practical perspective, the project demonstrates the application of Transaction Processing System (TPS) concepts in a real-world scenario.

It highlights how accurate and consistent transaction data can serve as the operational backbone of a business and act as a foundation for future decision-support and management systems.

In conclusion, the project successfully meets its defined objectives and proves that a well-designed e-commerce system can significantly enhance business operations, customer experience, and data reliability.

3.3 Recommendations

Although the current version of the system meets its intended goals, several improvements and enhancements can be considered for future development:

i. **Online Payment Gateway Integration**
   Integrating secure payment gateways such as eSewa, Khalti, or international payment services would enable real-time online payments and enhance user convenience.

ii. **Advanced Reporting and Analytics**
   The system can be extended into a full Management Information System (MIS) by adding sales reports, revenue analysis, customer behavior tracking, and graphical dashboards.

iii. **Inventory Management Module**
   Automatic stock updates, low-stock alerts, and supplier management features would improve operational efficiency and prevent stock shortages.

iv. **Enhanced Security Measures**
   Implementation of HTTPS, stronger password policies, multi-factor authentication, and role-based access control would further strengthen system security.

v. **Mobile Application Support**
   Developing a mobile-friendly or dedicated mobile application would improve accessibility and user engagement.

vi. **Scalability and Cloud Deployment**
   Deploying the system on scalable cloud platforms with optimized databases would allow it to handle higher traffic and transaction volumes.

These enhancements would significantly improve system functionality, usability, and long-term sustainability.

REFERENCES

Alter, S. (2022). Long-term value of transaction processing systems. *Information and Management, 59*(3), 103500.

Bakoben, N., Giudici, P., & Figini, S. (2020). Structured transactional data for prediction and operational planning. *Data Science Journal, 19*(1), 12–28.

Bhatt, R., Chennells, R., & Vause, M. (2023). Ensuring financial stability through robust transaction systems. *International Journal of Digital Business, 15*(1), 33–49.

Campbell, J. Y., Martinez-Jerez, F., & Tufano, P. (2022). Automated transaction systems and operational accuracy. *Journal of Financial Technology, 8*(3), 45–61.

Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital marketing: Strategy, implementation and practice* (7th ed.). Pearson.

Chen, H., Chiang, R., & Storey, V. (2019). Business intelligence and high-quality transactional data. *Journal of Information Systems, 33*(2), 85–101.

Dennis, A., Wixom, B., & Roth, R. (2021). System development methodologies for reliable TPS. *Systems Analysis and Design Journal, 19*(2), 120–138.

Heizer, J., Render, B., & Munson, C. (2020). *Operations management: Sustainability and supply chain management* (13th ed.). Pearson.

Jakubik, P., & Teleu, I. (2025). Historical transaction data supporting predictive analytics. *Journal of Business Analytics, 8*(1), 15–32.

Kumar, V., & Reinartz, W. (2018). Customer data management for personalization and engagement. *Journal of Marketing Analytics, 6*(2), 73–89.

Laudon, K. C. (2020). Automation and organizational efficiency through transaction systems. *MIS Quarterly Executive, 19*(3), 145–158.

Laudon, K. C., & Laudon, J. P. (2022). *Management information systems: Managing the digital firm* (17th ed.). Pearson.

Le, T., Nguyen, P., & Tran, H. (2023). Scalable transaction systems and digital platform growth. *Journal of E-Commerce Research, 24*(2), 101–118.

O'Brien, J., & Marakas, G. M. (2020). *Introduction to information systems* (17th ed.). McGraw-Hill Education.

Peffers, K., Tuunanen, T., Rothenberger, M., & Chatterjee, S. (2018). Design science research in information systems. *Journal of Management Information Systems, 24*(3), 45–77.

Satzger, G., Psaier, H., & Hummer, W. (2021). Modular design of e-commerce platforms for future adaptation. *Information Systems Journal, 31*(4), 567–589.

Turban, E., Pollard, C., & Wood, G. (2021). E-commerce transaction processing. In R. Laudon (Ed.), *Introduction to e-commerce* (pp. 112–135). Springer.

Wang, H. (2021). Centralized digital transaction systems and organizational transparency. *International Journal of Information Management, 56*, 102258.

West, D. M. (2021). *Digital transformation and information systems*. MIT Press.

Zwass, V. (2019). E-commerce systems as transaction-centric platforms. *Communications of the Association for Information Systems, 44*, 33–52.

APPENDICES

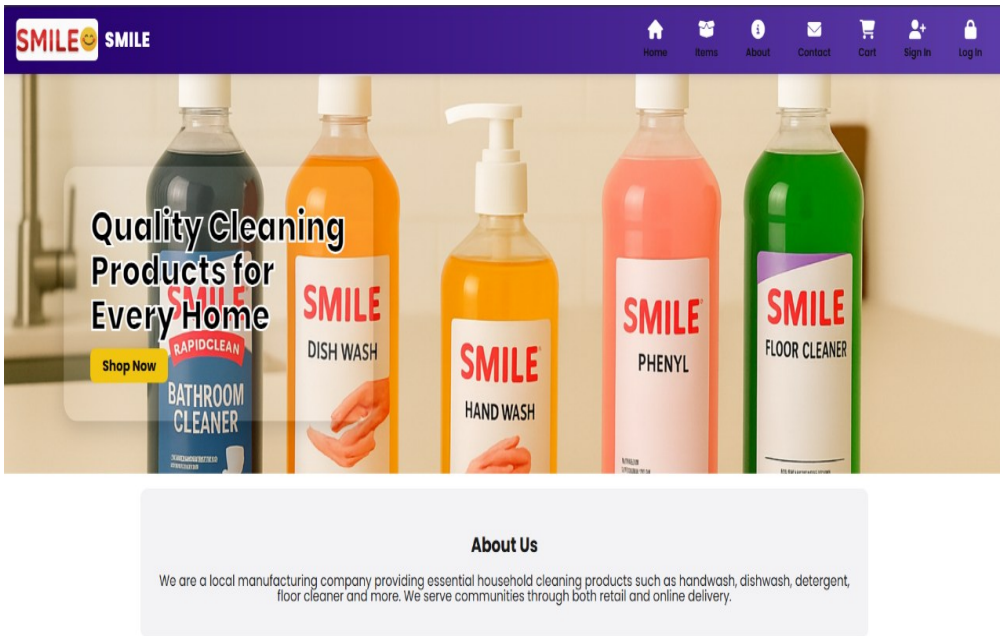

Figure A1: Homepage of the Website

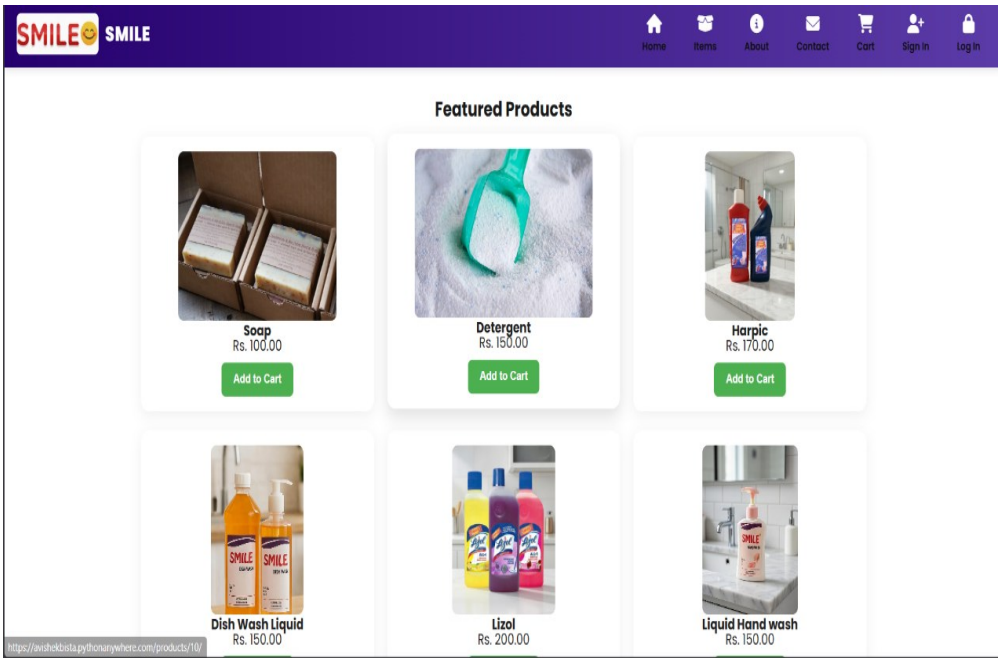

Figure A2: Product Listing Page

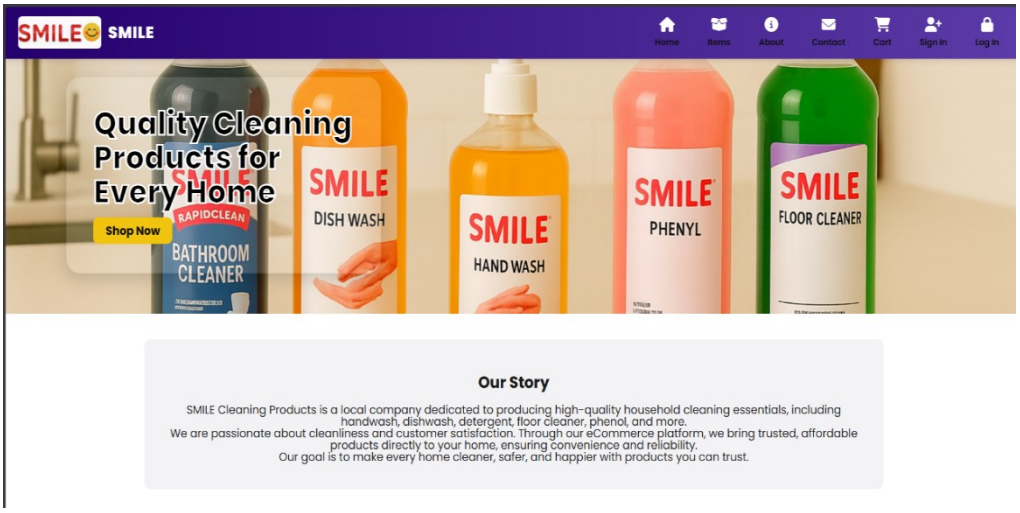

Figure A3: About Us Page

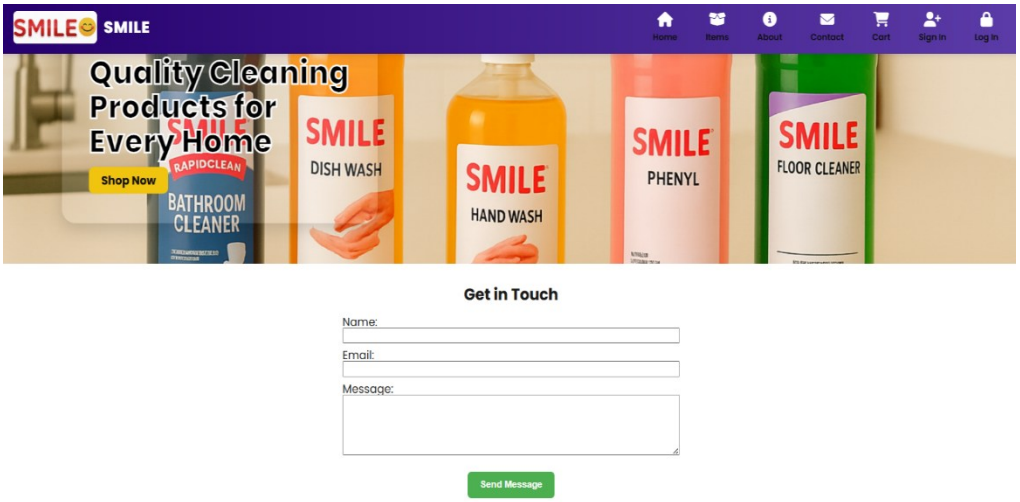

Figure A4: Contact Us Page

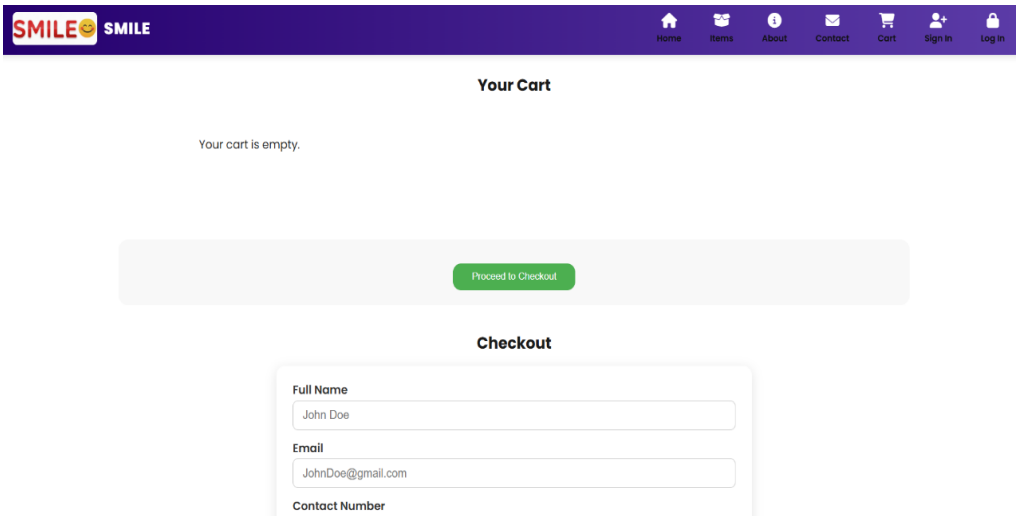

Figure A5: Cart and Checkout Page

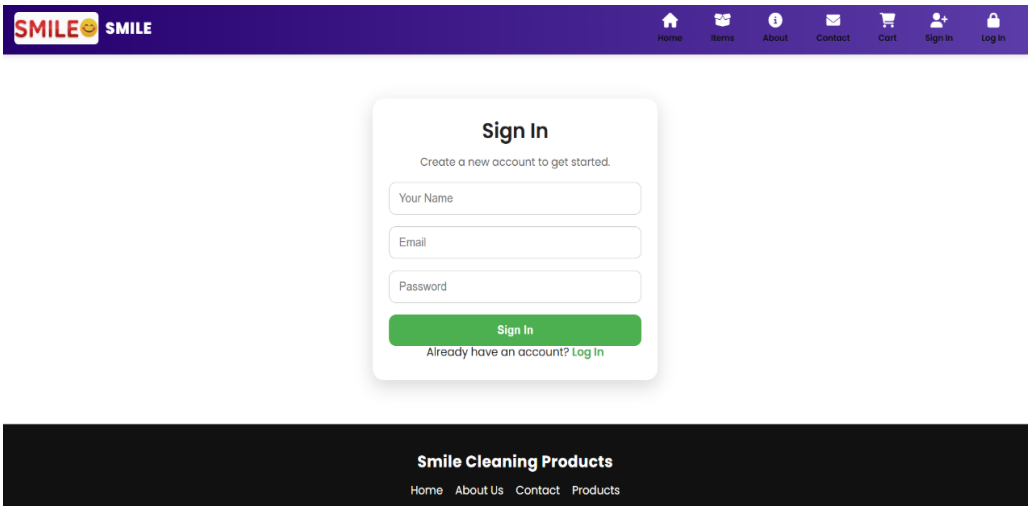

Figure A6: Sign In Page

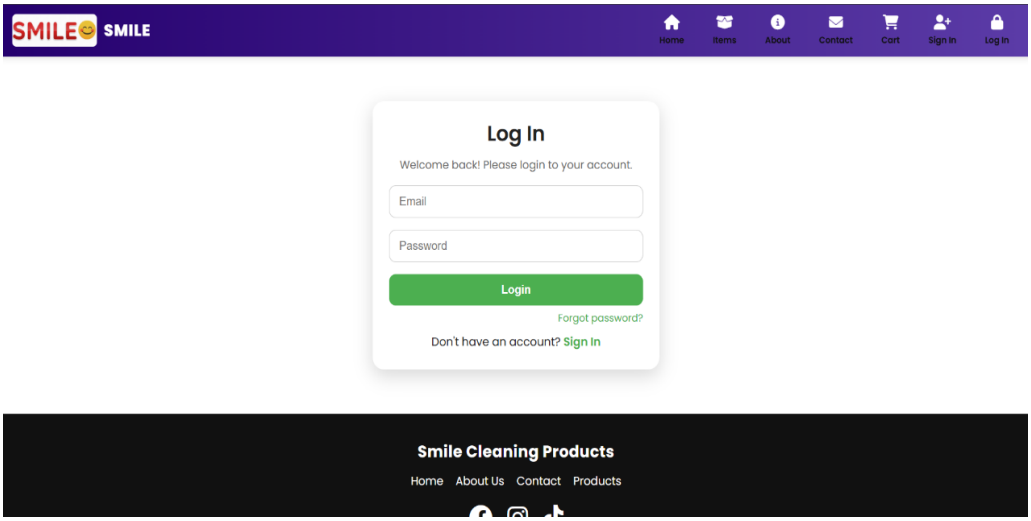

Figure A7: Log In Page